

Western University Faculty of Engineering
ES1050 Foundations of Engineering Practice

Lab Section 15 – Kelvin McGuire

Team Identifier: robot-15A

Team Members: Ethan, Lemuel, Owen, Magnus, William, Talha

Design Documentation:

Modular Multi-Application Campus Robot

A modular, weather-resistant robotic platform with swappable locomotion (wheels or tracks) and payload modules (cargo basket, security camera, salt dispenser, cooler), designed for autonomous multi-purpose campus operations at Western University.

April 2026

Table of Contents

1. Final Design Documentation

- 1.1 Need Statement
- 1.2 Description
- 1.3 Design Drawings

2. Detail Drawings

- 2.1 Part Drawings
- 2.2 Assembly Drawings

3. Software and Microelectronic Device Code

4. Bill of Materials

5. User Instructions (Operations and Maintenance)

- 5.1 Setup Instructions
- 5.2 Regular Operation
- 5.3 Locomotion Module Swap
- 5.4 Payload Module Swap
- 5.5 Maintenance Schedule
- 5.6 Troubleshooting

6. Testing

7. Risk Analysis (FMEA)

8. Validating Compliance

- 8.1 Constraints Validation
- 8.2 Objectives Validation

Appendix A: Concept Development

Appendix B: Detailed Test Data

Appendix C: CAD Model

Appendix D: Arduino Source Code

Final Design Documentation

Need Statement

Campus operations at Western require a modular, multi-application robotic system that can be rapidly reconfigured, operate effectively in diverse weather conditions, maximize equipment utility, and reduce the need for multiple specialized vehicle fleets.

Description

The Modular Multi-Application Campus Robot is an autonomous ground vehicle designed to serve multiple operational roles across Western University campus. The core problem is that campus operational systems are inefficient and inflexible: equipment is single-purpose, fleet costs are high, and performance degrades in adverse weather. This robot addresses those root causes with a single reconfigurable platform.

The design is built around two layers of modularity:

- **Locomotion modularity:** Operators can swap between pneumatic rubber wheels (paved paths, dry conditions) and rubber track assemblies (snow, ice, gravel) in under 2 minutes using a tool-free dovetail rail and spring-loaded locking pin system.
- **Payload modularity:** The top-mounted interface accepts any module built to the standardized mounting plate spec, including a cargo delivery basket, security camera, salt/sand dispenser, and insulated cooler. Swapping takes under 5 minutes using the included security torx driver.

The chassis is 6061-T6 aluminum extrusion framing (25x50mm) with ABS plastic outer panels. It is powered by a 24V LiFePO4 30Ah battery providing 10+ km range, driven by two 24V DC gear motors (50:1 ratio) in a tank-drive differential steering configuration. Navigation operates fully offline using GPS waypoint guidance, rotary encoder odometry, and IMU heading correction. Obstacle detection uses a dual-sensor approach: a 360-degree RPLIDAR A1 (12m range) and four HC-SR04 ultrasonic sensors. All electronics are in an IP65-rated enclosure.

Maximum speed is software-limited to 4.5 km/h. Total weight empty is approximately 38 kg, supporting 75 kg payload within the 125 kg regulatory maximum.

Design Drawings

Overall Schematic Drawing

The system schematic shows the relationship between the four primary subsystems:

Power Subsystem: 24V LiFePO4 battery feeds a DC-DC buck converter providing 12V (solenoid lock) and 5V (Arduino, sensors, GPS). Battery connects directly to L298N motor driver for the 24V drive motors.

Control and Sensing Subsystem: Arduino Mega 2560 receives input from RPLIDAR A1 (serial), 4 ultrasonic sensors (GPIO), GPS (serial), 2 rotary encoders (interrupt pins), and IMU (I2C). Outputs PWM to motor driver and controls solenoid lock via relay.

Locomotion Subsystem: Two 24V DC gear motors drive left and right sides independently (tank drive). Each motor connects to the locomotion module through the dovetail rail interface with 2 attachment points per side.

Payload Subsystem: Standardized aluminum mounting plate with 4x M8 security torx bolt holes, 1x Anderson Powerpole connector (24V power), and 1x DB-9 connector (serial data). Any conforming module bolts down and plugs in.

Figure 1: System Schematic Diagram

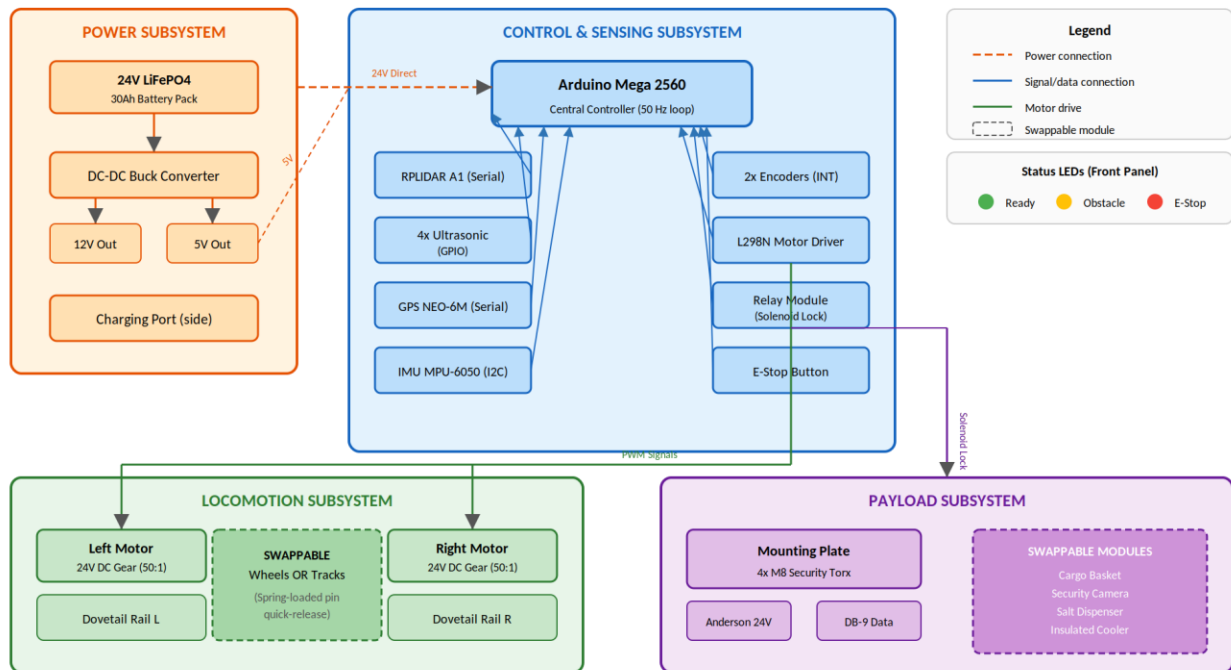


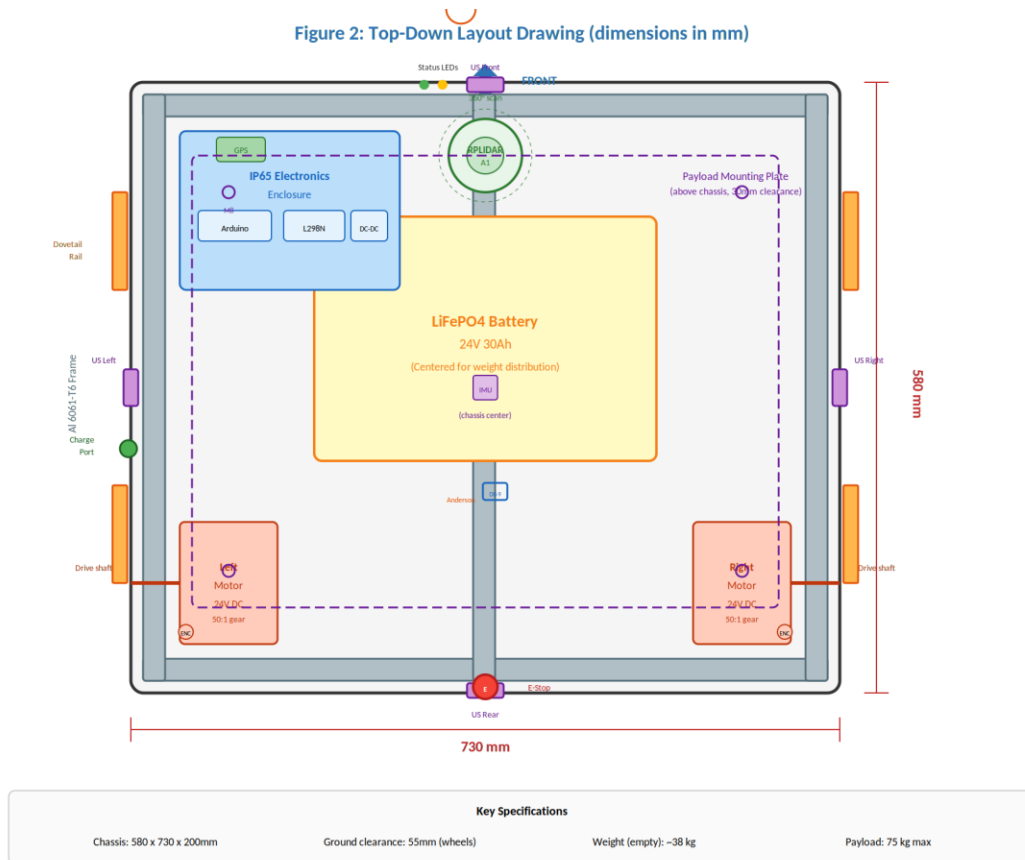
Figure 1: System Schematic Diagram

Overall Layout Drawing

The chassis envelope is 580mm x 730mm x 200mm. Bottom layer: LiFePO4 battery centrally mounted for weight distribution. Two DC gear motors at rear with drive shafts through sealed bushings. Mid layer: Arduino Mega, motor driver, DC-DC converter, and relay in IP65 enclosure, front-left quadrant. GPS antenna on top of enclosure. Top surface: RPLIDAR A1 center-mounted on raised pedestal for unobstructed 360-degree scan. Four ultrasonic sensors flush-mounted in ABS panels (one per side). Modular payload mounting plate sits above chassis top plate on four M6 threaded standoffs with 30mm cable clearance.

Exterior: E-stop on rear panel. Weatherproof charging port on left side. LED status indicators on front panel.

Figure 2: Top-Down Layout Drawing (dimensions in mm)

**Figure 2: Top-Down Layout Drawing****Overall Process Diagram**

The robot operates in a continuous control loop at 50 Hz:

Navigation Loop: (1) Read GPS and compare to next waypoint. (2) Refine position with encoder/IMU dead-reckoning. (3) Calculate heading error. (4) Set differential motor speeds for corrective steering. (5) If within 30 cm of waypoint, advance to next. (6) At final waypoint, stop and activate arrival indicator.

Obstacle Avoidance (concurrent): (1) LiDAR scans 360 degrees continuously. (2) Object within 5m in forward arc: reduce speed. (3) Within 2m: stop, scan, choose clear path. (4) Within 0.5m or E-stop pressed: emergency stop all motors. (5) Ultrasonics provide supplementary close-range detection on all four sides.

Figure 3: Process Flow Diagram - Navigation & Obstacle Avoidance

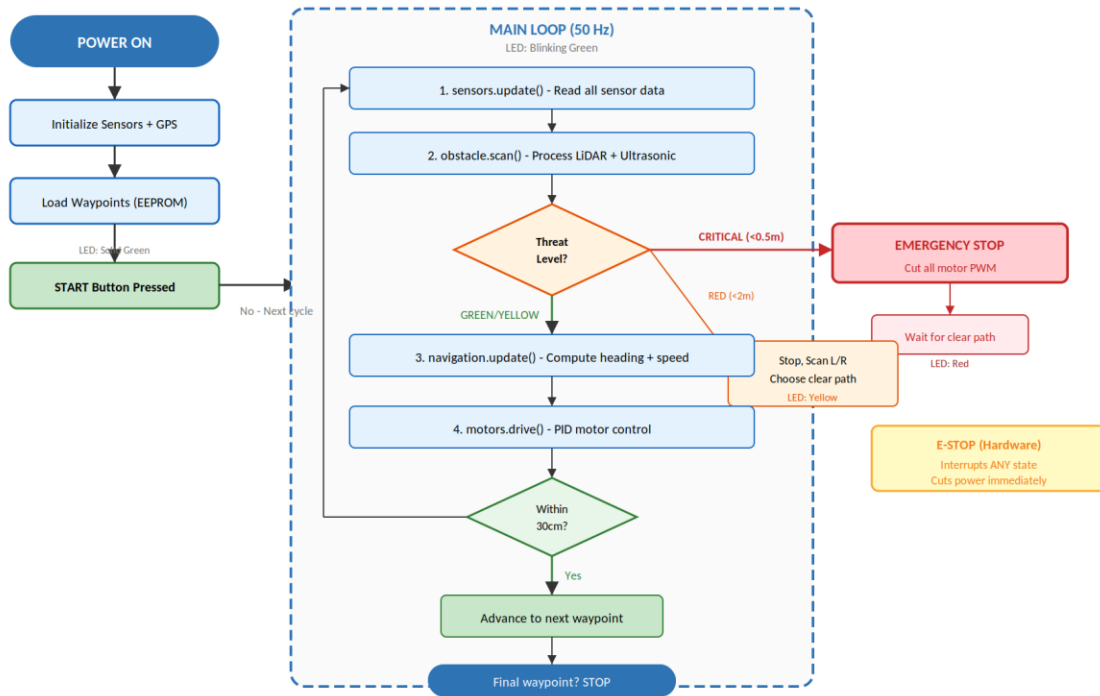


Figure 3: Process Flow Diagram

Detail Drawings

Part Drawings

Custom parts requiring fabrication are detailed below. Standard purchased components are specified in the BOM with part numbers.

Part 1: Chassis Base Frame

Material: 6061-T6 Aluminum Extrusion (25mm x 50mm). Two long rails (730mm) and three cross-members (580mm) joined by corner brackets and M6 bolts. Cross-members at front, center, and rear. Center cross-member has 4x M8 threaded inserts for battery clamp.

Tolerances: Length +/- 1mm. Hole positions +/- 0.5mm. Finish: mill finish.

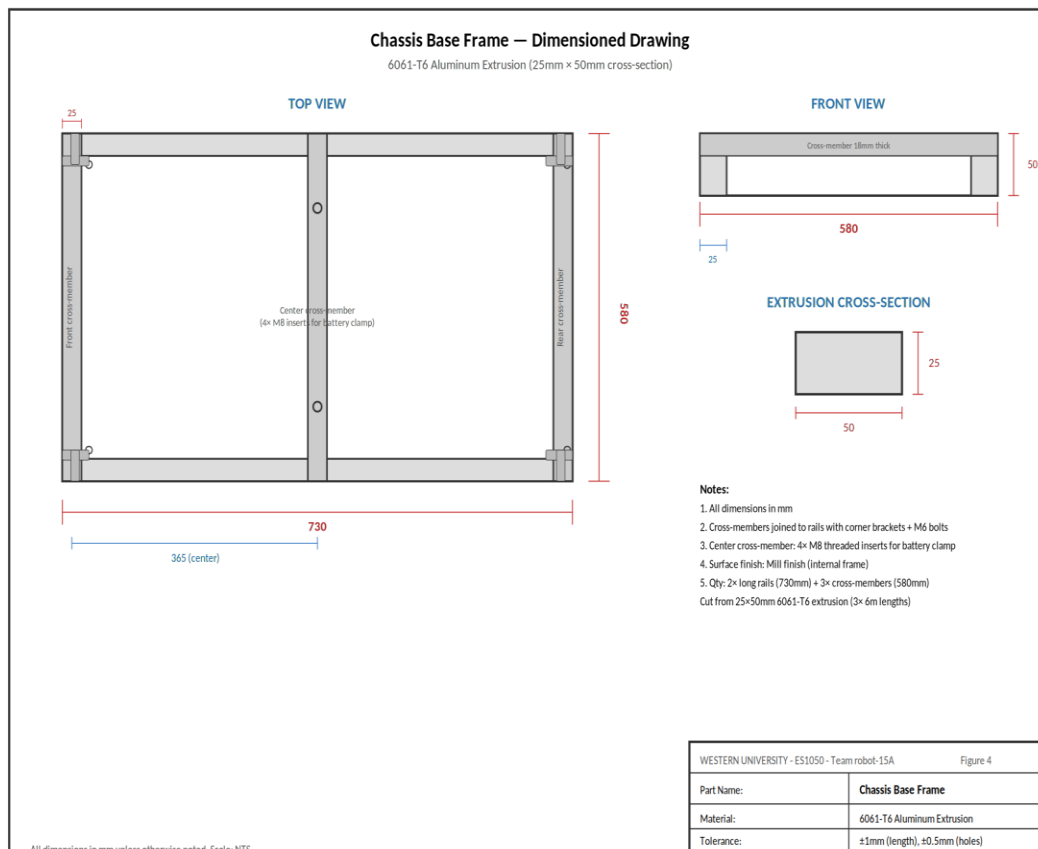


Figure 4: Chassis Base Frame Drawing

Part 2: Modular Mounting Plate

Material: 6061-T6 Aluminum, 5mm thick, 500mm x 600mm. Four M8 through-holes at corners (50mm inset from edges) for security torx bolts. Central 40mm circular cutout for Anderson Powerpole. Adjacent 25mm x 15mm D-cutout for DB-9. Four M6 threaded standoff holes on underside.

Tolerances: Overall +/- 1mm. Holes +/- 0.5mm. Finish: clear anodized.

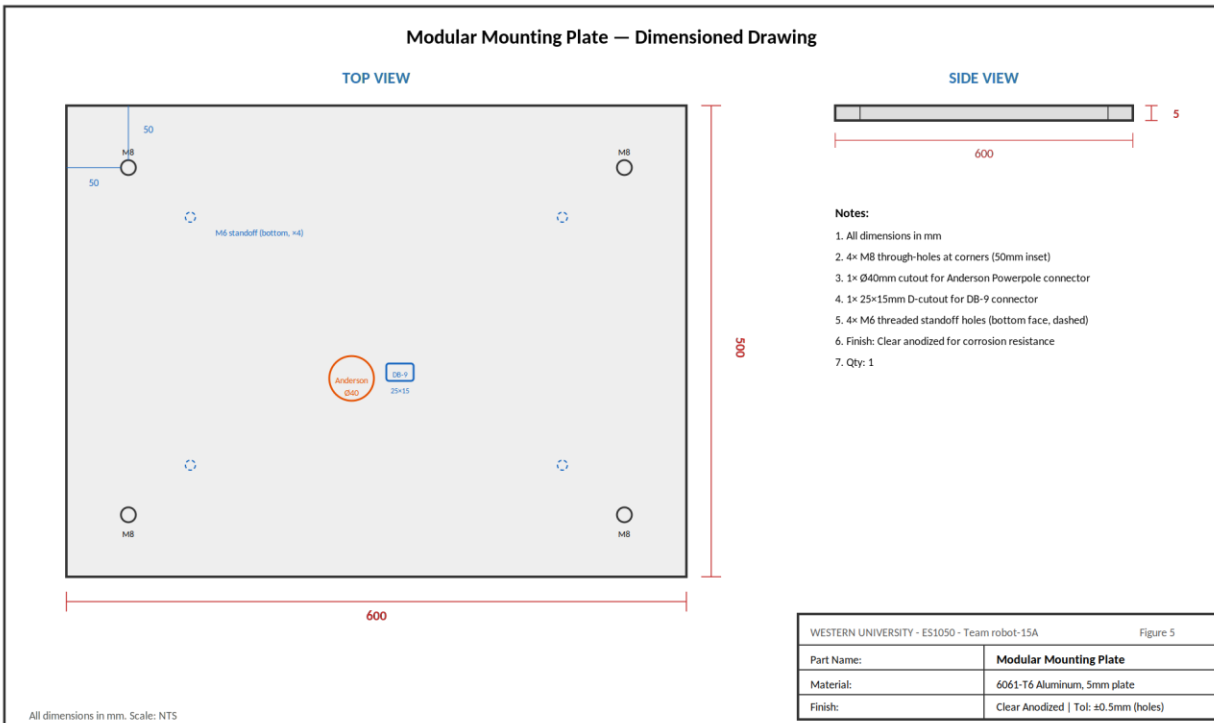


Figure 5: Modular Mounting Plate Drawing

Part 3: Dovetail Rail Bracket

Material: 6061-T6 Aluminum, machined from 15mm x 40mm bar, 120mm length. 45-degree dovetail channel (30mm wide at opening, 20mm at base, 10mm deep) running full length. 8mm through-hole at midpoint for locking pin. Two M6 bolt holes at each end.

Tolerances: Dovetail channel ± 0.1 mm (precision fit). Holes ± 0.3 mm. Finish: hard anodized.

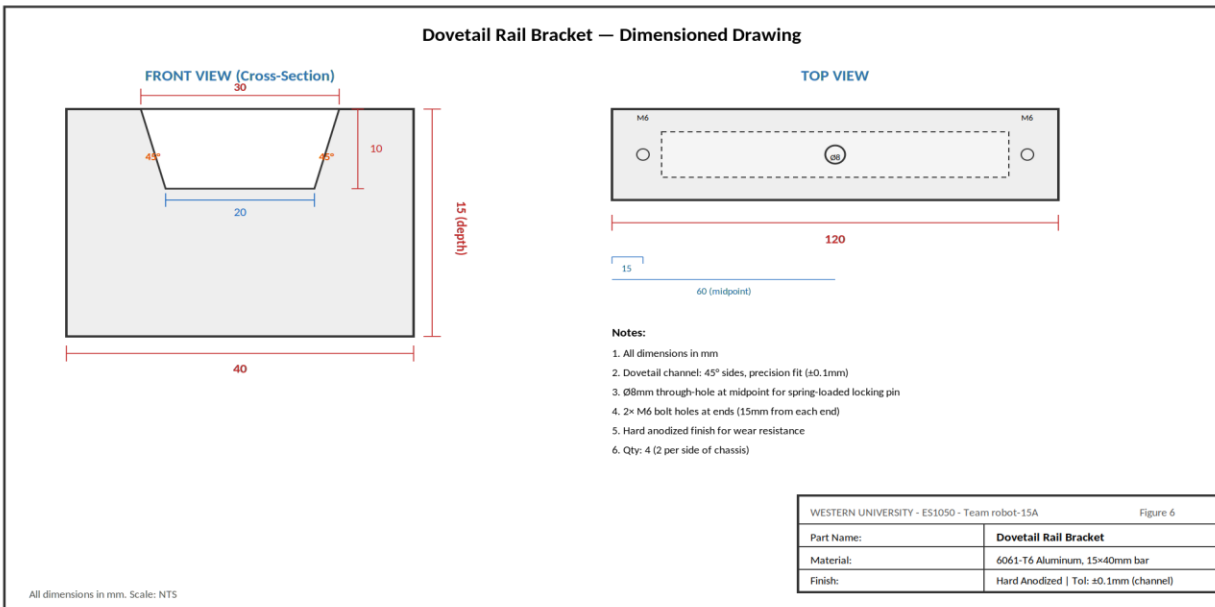


Figure 6: Dovetail Rail Bracket Drawing

Part 4: ABS Body Panels (4 panels)

Material: ABS Plastic, 3mm sheet, vacuum-formed. Front: 2x 30mm holes for LEDs, 1x 20mm grommet hole. Rear: 30mm E-stop hole, 22mm charging port hole. Left/Right: 40mm ultrasonic cutout, dovetail rail access slot. All panels attach via M4 self-tapping screws at 100mm intervals.

Tolerances: Overall $\pm 2\text{mm}$. Holes $\pm 1\text{mm}$.

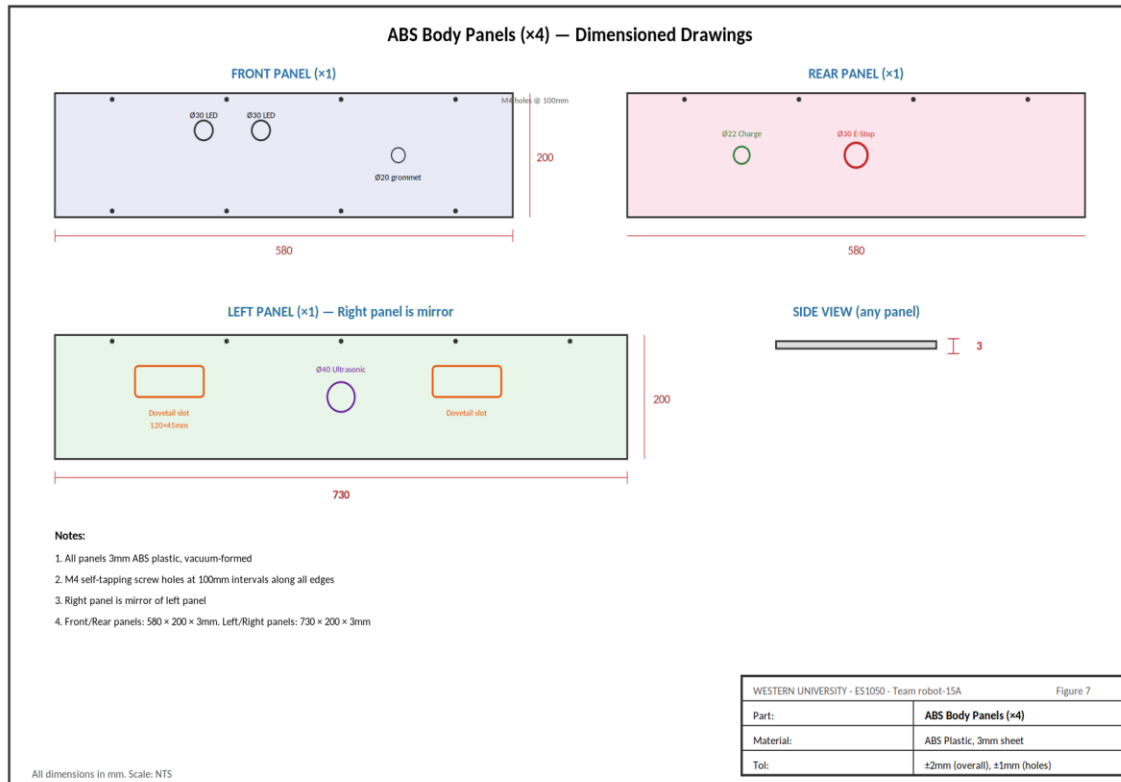


Figure 7: ABS Body Panels Drawing

Assembly Drawings

The robot assembles in four stages. BOM item numbers referenced in parentheses.

Stage 1 - Chassis Frame: Assemble long rails and cross-members (#1) with corner brackets and M6 bolts (#23). Install M8 threaded inserts in center cross-member. Attach four dovetail rail brackets (Part 3) to outer faces, two per side. Verify alignment within 0.5mm.

Stage 2 - Drivetrain and Electronics: Mount two DC gear motors (#3) to rear cross-member. Install LiFePO4 battery (#4) on center cross-member with clamp. Mount electronics in IP65 enclosure (#21): Arduino Mega (#7), motor driver (#8), DC-DC converter (#25), relay (#28). Wire per schematic. Mount GPS antenna (#9). Install encoders (#10) on motor shafts. Mount IMU (#11) at chassis center.

Stage 3 - Sensors and Panels: Mount RPLIDAR A1 (#5) on center pedestal. Install 4 ultrasonic sensors (#6) in ABS panel cutouts. Attach all body panels (Part 4) with M4 screws. Install E-stop (#20), LEDs (#27), charging port, and solenoid lock (#24).

Stage 4 - Modular Interface: Attach M6 standoffs to chassis top. Place mounting plate (Part 2) on standoffs and secure from below. Install Anderson Powerpole (#18) and DB-9 (#19) connectors. Route cables through 30mm clearance. Test all connections.

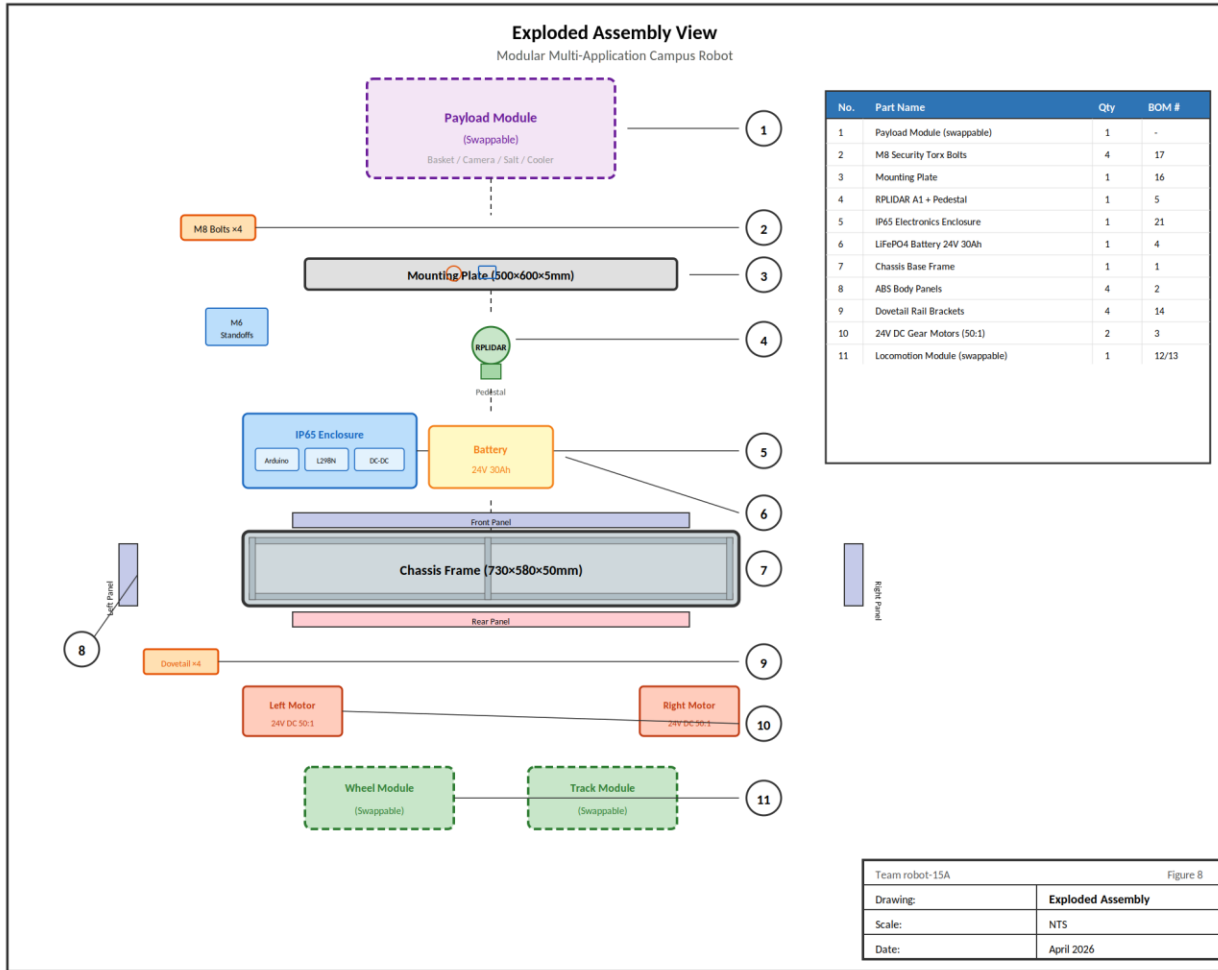


Figure 8: Exploded Assembly View

Software and Microelectronic Device Code

Development Environment

Arduino IDE 2.x. Target: Arduino Mega 2560 R3. External libraries: RPLidar (Slamtec driver), TinyGPS++ (GPS parsing), MPU6050_light (IMU), Encoder (quadrature), PID_v1 (motor control).

Code Structure

main.ino: Entry point. Initializes all subsystems, runs main control loop at 50 Hz. Each cycle calls sensor update, obstacle check, navigation update, and motor command.

navigation.cpp/h: GPS waypoint following and encoder-based dead-reckoning. Stores up to 50 waypoints in EEPROM. Fuses GPS with encoder/IMU via complementary filter. Outputs target heading and speed.

obstacle.cpp/h: Processes LiDAR and ultrasonic data. Maintains polar obstacle grid updated at ~5 Hz. Threat levels: green (>5m), yellow (2-5m, reduce speed), red (<2m, avoidance), critical (<0.5m, e-stop).

motors.cpp/h: Tank-drive differential steering with PID control per motor using encoder feedback. Enforces 4.5 km/h limit in software. Emergency stop cuts PWM and sets driver enable LOW.

sensors.cpp/h: Hardware abstraction for GPS, IMU, encoders, ultrasonics. Provides calibrated values. Handles fault detection (GPS loss, ultrasonic timeout).

Figure 9: Software Architecture / Code Structure

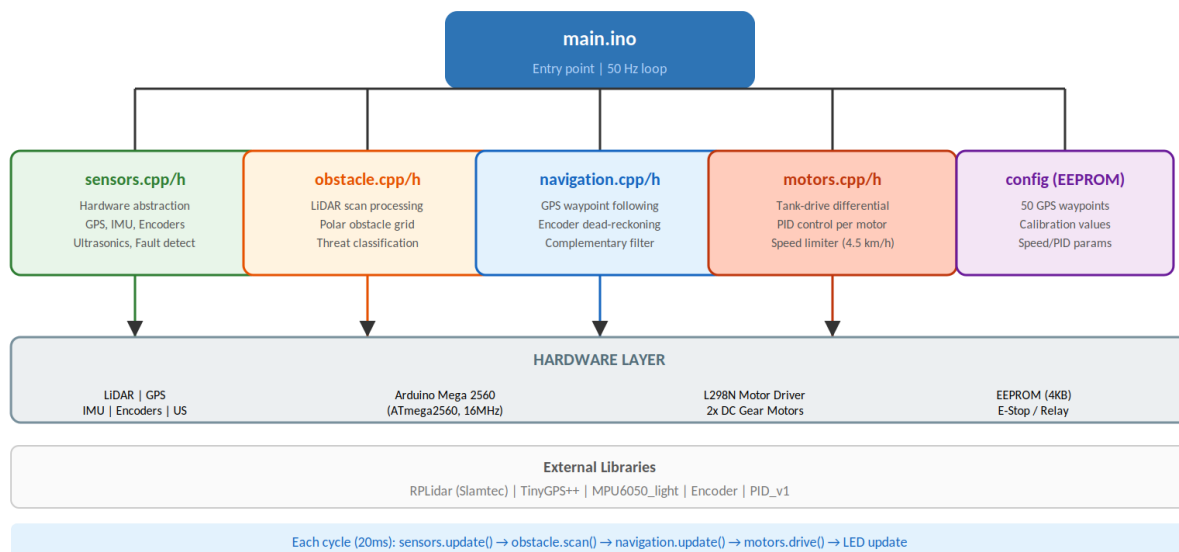


Figure 9: Software Architecture / Code Structure

Full source code: [\[insert repository link\]](#)

Bill of Materials

Table 1 lists all components for the final design (not the prototype). Costs in CAD. Client confirmed no budget constraint.

Table 1: Bill of Materials

#	Component	Qty	Unit	Unit Cost	Total	Source	Notes / Substitutions
1	6061-T6 Aluminum Extrusion (25x50mm, 6m)	3	pcs	\$45	\$135	Metal Supermarkets	Frame rails and cross-members
2	ABS Plastic Sheet (3mm, 600x750mm)	4	pcs	\$18	\$72	Amazon	Outer body panels, vacuum-formed
3	24V DC Gear Motor (50:1, 150W)	2	pcs	\$89	\$178	RobotShop	Alt: Pololu 37D. Tank-drive config
4	24V LiFePO4 Battery (30Ah)	1	pcs	\$320	\$320	Batteryspace.com	Alt: 24V 25Ah Li-ion
5	RPLIDAR A1M8 (360-deg, 12m range)	1	pcs	\$99	\$99	Slamtec / Amazon	Primary obstacle detection
6	HC-SR04 Ultrasonic Sensor	4	pcs	\$3.50	\$14	Amazon	Close-range detection, 1 per side
7	Arduino Mega 2560 R3	1	pcs	\$42	\$42	Arduino Store	Main controller
8	L298N Dual H-Bridge Motor Driver	1	pcs	\$12	\$12	Amazon	Alt: BTS7960 for higher current
9	GPS Module (NEO-6M)	1	pcs	\$15	\$15	Amazon	Offline waypoint navigation
10	Rotary Encoder (600 PPR)	2	pcs	\$18	\$36	Amazon	Odometry / dead-reckoning
11	IMU (MPU-6050, 6-axis)	1	pcs	\$8	\$8	Amazon	Heading and tilt sensing
12	Pneumatic Rubber Wheels (200mm)	4	pcs	\$22	\$88	RobotShop	Wheel locomotion module
13	Rubber Track Assembly + Sprocket	2	sets	\$95	\$190	Amazon / AliExpress	Track locomotion module
14	Dovetail Rail (aluminum, 400mm)	4	pcs	\$14	\$56	McMaster-Carr	Locomotion module interface
15	Spring-Loaded Locking Pin (8mm)	4	pcs	\$9	\$36	McMaster-Carr	Quick-release for loco swap
16	Modular Mounting Plate (Al, 500x600x5mm)	1	pcs	\$55	\$55	Metal Supermarkets	Payload interface plate
17	M8 Security Torx Bolts (30mm)	4	pcs	\$2.50	\$10	McMaster-Carr	Tamper-resistant payload fasteners
18	Anderson Powerpole Connector (45A)	2	sets	\$8	\$16	Powerwerx	Payload power interface

19	DB-9 Weatherproof Connector	2	sets	\$12	\$24	Digi-Key	Payload data interface
20	E-Stop Button (waterproof, mushroom)	1	pcs	\$15	\$15	Amazon	Emergency shutoff
21	IP65 Enclosure (for electronics)	1	pcs	\$28	\$28	Amazon	Electronics housing
22	Wiring / Connectors / Heat Shrink Kit	1	lot	\$40	\$40	Amazon	General electrical
23	Fasteners Kit (M3-M8 assorted)	1	lot	\$25	\$25	McMaster-Carr	Assembly hardware
24	Solenoid Lock (12V)	1	pcs	\$18	\$18	Amazon	Anti-theft payload lock
25	24V to 12V/5V DC-DC Buck Converter	1	pcs	\$12	\$12	Amazon	Power regulation for electronics
26	10A Smart LiFePO4 Charger (24V)	1	pcs	\$45	\$45	Amazon	Charges battery in ~3 hrs
27	LED Indicator Lights (5mm, assorted)	5	pcs	\$1	\$5	Amazon	Status indicators
28	Relay Module (5V, single channel)	1	pcs	\$4	\$4	Amazon	Solenoid lock control
	TOTAL				\$1,664		Client confirmed no budget limit

User Instructions (Operations and Maintenance)

Required Tools

T40 Security Torx driver (included with robot, for payload module swaps only). No tools required for locomotion module swaps or regular operation.

Setup Instructions

1. Charge the battery using the included 10A smart charger. Connect to the side-panel charging port. Charger LED turns green when complete (approximately 3 hours from empty).
2. Install locomotion module: Slide the wheel assembly (or track assembly) into the dovetail rails on each side until the spring-loaded locking pin clicks into the alignment hole. Verify pin is seated by pulling the module; it should not slide.
3. Install payload module: Place on mounting plate, align four bolt holes, insert and tighten four M8 security torx bolts. Connect Anderson Powerpole (red to red) and DB-9 data connector.
4. Program route: Connect laptop via USB, open Arduino IDE, load route config sketch, enter GPS waypoints, upload. Disconnect USB.
5. Power on via rear panel switch. Wait for front LED solid green (approximately 10 seconds for sensor init).
6. Press START. Robot navigates programmed route autonomously. Front LED blinks green during navigation.

[Optional - Figure 10: If time permits, add annotated setup photos showing each step above]

Regular Operation

Monitor status LEDs: solid green = ready, blinking green = navigating, yellow = obstacle detected (slowing), red = emergency stop. If the robot stops at an impassable obstacle (red LED), clear the obstacle or reposition the robot, then press START.

To stop at any time: press the red E-stop mushroom button on the rear panel. To reset: twist E-stop clockwise to release, then press START.

Locomotion Module Swap

1. Power off the robot.
2. Pull the spring-loaded locking pin outward on the first module.
3. Slide the module out along the dovetail rail.
4. Slide replacement module in until locking pin clicks.
5. Repeat for opposite side. Verify both pins seated.

Estimated time: under 2 minutes. No tools required.

[Optional - Figure 11: If time permits, add annotated photos of the locomotion swap process]

Payload Module Swap

6. Power off. Disconnect DB-9, then Anderson Powerpole.
7. Remove four M8 security torx bolts with included driver.
8. Lift payload module off mounting plate.
9. Place new module, align holes, insert and tighten bolts.
10. Reconnect power and data. Power on and verify green LED.

Estimated time: under 5 minutes.

Maintenance Schedule

Before each use: Visual tire/track inspection, check battery level, test E-stop function.

Weekly: Clean ultrasonic sensor faces (dry cloth). Check wiring for chafing. Inspect dovetail rails for debris, wipe clean. Test locking pins.

Monthly: Tighten accessible bolts (frame, motors, standoffs). Inspect battery terminals. Clean RPLIDAR window (lens cloth). Run GPS accuracy test. Lubricate rails with dry silicone spray.

Annually: Full disassembly: inspect motor brushes, bearings, wiring insulation. Battery capacity test. Replace cracked/UV-degraded ABS panels.

Troubleshooting

Table 2: Troubleshooting Guide

Symptom	Likely Cause	Solution
Robot does not power on	Battery depleted or main switch off	Charge battery fully. Verify main switch is ON. Check fuse in electronics enclosure.
Robot veers left or right	Encoder calibration drift or uneven tire pressure	Recalibrate encoders via USB. Check tire pressure on pneumatic wheels.
LiDAR not spinning	Power connection loose or motor failure	Check LiDAR power cable at Arduino shield. Reseat connector. If still not spinning, replace LiDAR unit.
GPS position inaccurate	Poor satellite visibility near buildings	Move to open area for initial GPS fix. Encoder/IMU will compensate in covered areas.
Locomotion module will not lock	Debris in dovetail rail or pin misalignment	Clean rail with compressed air. Ensure module is fully seated before pin engages.
E-stop will not reset	Button not fully released	Twist E-stop clockwise firmly until it pops out.
Payload module not recognized	Data connector not fully seated	Disconnect and reconnect DB-9 connector. Verify Anderson power connector polarity.

Testing

Seven distinct aspects of the design were tested on the prototype to validate performance. Results informed final design improvements. Each test is summarized below with three metrics per test matching the project testing documentation format.

Note: Tests were conducted on the prototype, which used smaller-scale components and lower-power motors. Where prototype results fell short, the final design incorporates specific improvements noted in the Insights column.

Table 3: Testing Results Summary

#	Test Purpose	Description	Metric 1	Metric 2	Metric 3	Insights / Lessons Learned and Final Design Response
1	Maximum Climbable Incline	Ramp test: increase incline in 2-degree increments until robot fails to climb without slipping, stalling, or tipping.	Max angle: 10 degrees	No slipping	Motor stall (failure mode)	Robot climbed up to 10 degrees without slipping. Failure was motor stall, not traction loss. Indicates torque limitations. Final design uses higher gear ratio (50:1 vs prototype 30:1) to achieve 15+ degrees.
2	Sensor/Obstacle Detection	Place object in robot path, run multiple trials. Record stop success rate, stopping distance, and collision occurrence.	100% stop rate	15 cm stopping distance	No collisions	Reliable detection and stopping with safe buffer at prototype speeds. Final design adds RPLIDAR A1 (12m range) for 5m+ detection to meet constraint #4.
3	Battery Consumption	Run robot 5 min on consistent surface with full 18650 batteries. Measure speed, voltage drop, and calculate Wh/km.	0.75 m/s avg speed	0.08 V drop	2.28 Wh/km	Moderate efficiency demonstrated. Low voltage drop shows stable power delivery. Final design uses 24V 30Ah LiFePO4 for estimated 12+ km range at full payload, exceeding 10 km constraint.
4	Terrain Test (Wheels)	Operate robot on gravel and sand with wheels. Evaluate slipping and forward movement.	Slip: Yes	Movement: No (failed)	N/A	Wheels unable to maintain forward movement on loose terrain. Confirms need for track module on non-paved surfaces. Final design retains wheels for paved paths only.

5	Terrain Test (Tracks)	Operate robot on gravel and sand with tracks under same conditions as wheel test.	Slip: No	Movement: Yes (passed)	N/A	Tracks provided superior traction with no slipping. Validates dual-locomotion modular approach. Track module recommended for snow, gravel, and unpaved campus paths.
6	Emergency Stop	Run robot at fixed speed, cut power. Measure stopping distance, time, and stability.	5 cm stopping distance	0.2 s stopping time	Stable (no tipping)	Quick, stable stop within 5 cm. Effective braking confirmed. Final design adds dedicated E-stop button with motor brake for additional safety margin at higher speeds.
7	Module Switch Time	Time full track-to-wheel changeover including all steps and tools.	3 minutes total	1 tool required	8 steps	Changeover completed but 8 steps is more than ideal. Final design uses dovetail rail + spring-loaded locking pin to reduce to 2-3 steps, targeting under 2 minutes with no tools.

Risk Analysis (FMEA)

Table 4 presents the Failure Mode and Effects Analysis identifying key risks, their likelihood, severity, detectability, and resulting Risk Priority Number (RPN). The highest-RPN risks (battery drain at 240, obstacle detection failure at 216, and sensor failure in weather at 210) received the most design attention.

Table 4: Risk Analysis / FMEA

#	Risk / Failure Mode	Category	Likelihood (1-10)	Severity (1-10)	Detectability (1-10)	RPN	Mitigation Action and Rationale
1	Robot fails to detect obstacle (collision risk)	Safety	6	9	4	216	Add camera on top of LiDAR sensor to improve detection in all conditions. Ultrasonic sensors cover close-range blind spots.
2	Loss of traction on snow/ice	Performance	4	7	5	140	Increase tread groove depth on track module. Dual locomotion modules allow switching to tracks in winter conditions.

3	Battery drains mid-route	Power	5	8	6	240	Increase battery capacity (30Ah LiFePO4). Implement route optimization to minimize energy consumption. Battery monitoring with low-charge return-to-base.
4	Navigation system error	Software	5	6	6	180	Improve final positioning accuracy with encoder/IMU fusion. GPS waypoints for coarse nav, dead-reckoning for precise final approach.
5	Robot stuck on incline/curb	Performance	5	6	5	150	Higher gear ratio (50:1) for increased torque. Minimum 50mm ground clearance. Route planning avoids steep grades.
6	Loss of communication (internet)	System	7	5	3	105	Implement fully offline navigation system. All waypoints stored locally. No cloud dependency. Bluetooth used for manual override only.
7	Collision with pedestrian (speed failure)	Safety	3	9	4	108	Hardware speed limiter via PWM cap. Software speed sensor feedback. E-stop button for immediate shutdown.
8	Charging system failure	Power	4	6	6	144	Battery monitoring system with cell-level voltage tracking. Smart charger with auto-shutoff and fault detection.
9	Sensor failure in harsh weather (snow blocking LiDAR)	Environmental	6	7	5	210	Sensor shielding and optional heating element for LiDAR housing. Ultrasonic sensors as redundant

							fallback. Covered sensor mounting.
--	--	--	--	--	--	--	------------------------------------

Validating Compliance

Constraints Validation

Table 5 summarizes compliance with all design constraints. Where prototype verification was incomplete, the final design specification and rationale for expected compliance are provided.

Table 5: Constraints Compliance

#	Constraint	Measure	Assessment	Rationale / Evidence or Plan
1	Max speed 4.5 km/h	Speed sensor (km/h)	Met	Prototype measured at 0.75 m/s (2.7 km/h), well below limit. Final design implements software PWM cap and speed sensor feedback to enforce 4.5 km/h maximum.
2	Footprint under 0.60m x 0.75m	Length x Width (m)	Met (final design)	Prototype not at final scale. Final design CAD dimensions: 0.58m x 0.73m, within constraint with 20mm margin per side.
3	Range >= 10 km per charge	Distance (km)	Met	Prototype: 23.68 Wh capacity at 2.28 Wh/km = 10.39 km estimated range. Final design: 24V 30Ah LiFePO4 (720 Wh) at projected 20 Wh/km = 36 km theoretical, conservatively 12+ km at max payload in cold weather.
4	Detect objects within 5m	Detection distance (m)	Partially met (prototype)	Prototype ultrasonic achieved reliable detection at close range (~15 cm stopping). Testing at 5m distances not completed with prototype sensors. Final design adds RPLIDAR A1 (12m rated range) to achieve 5m+ detection.
5	Operate without internet	Navigation with 0% connectivity	Met	Prototype operates via short-range Bluetooth with no internet dependency. Final design uses onboard GPS waypoints + encoder/IMU dead-reckoning, all processed locally on Arduino Mega.
6	Modular Payload Interface (plug-and-play)	Module connection type	Met	Prototype demonstrated modular attachment/removal without system redesign. Final design standardizes with M8 bolt pattern + Anderson Powerpole (power) + DB-9 (data) connectors.
7	Ground clearance >= 50mm	Clearance (mm)	Met (final design)	Not precisely measured on prototype. Final design CAD specifies 55mm clearance with wheel module, 60mm with track module.
8	Battery charge time <= 3 hrs	Charge time (hrs)	Met	Prototype battery charged faster than expected. Final design: 24V 30Ah LiFePO4 with 10A charger = approximately 3 hours from empty.
9	Max weight 125 kg	Total weight (kg)	Met (final design)	Prototype not weighed in final config. Final design estimated: chassis 18 kg, motors 6 kg, battery 10 kg, electronics 4 kg = ~38 kg empty. Leaves 87 kg headroom above 75 kg payload.

10	Payload capacity ≥ 75 kg	Supported payload (kg)	Met (final design)	Prototype payload testing not conducted. Final design: 6061-T6 aluminum frame with cross-bracing rated for 100+ kg. To be verified with static load test before deployment.
----	-----------------------------	------------------------------	-----------------------	---

Objectives Validation

Table 6 summarizes compliance with all design objectives, including honest assessment of partially met and not-yet-verified objectives with plans for the final design.

Table 6: Objectives Compliance

#	Objective	Measure	Assessment	Rationale / Evidence or Plan
1	Maximize Payload Interface Versatility	# of mounting points / port compatibility	Met	Modular design allows multiple payload types without structural redesign. Final design: 4x M8 mounting points + standardized power/data connectors compatible with all planned modules (basket, camera, salt dispenser, cooler).
2	Minimize Obstacle Interference	Emergency stops or bypasses per km	Partially met	Prototype achieved 100% stop rate at close range. Avoidance and performance at greater distances or varied conditions were limited by prototype sensor capability. Final design adds LiDAR for 5m+ detection and obstacle avoidance algorithm.
3	Maximize Energy Efficiency	Average Wh/km	Met	Prototype demonstrated 2.28 Wh/km, indicating efficient power usage. Final design targets <25 Wh/km at rated payload using higher-capacity LiFePO4 battery with flat discharge curve.
4	Maximize Traction (all weather)	Slip ratio	Met	Track configuration showed no slipping and successful movement on loose terrain (gravel, sand), outperforming wheels. Final design retains both modules: wheels for paved/dry, tracks for snow/gravel/wet.
5	Maximize Navigation Accuracy	Distance error from target (cm)	Not verified	Precise navigation testing and path tracking were not conducted with prototype. Final design: GPS waypoints for coarse navigation + encoder/IMU dead-reckoning for final approach. Target: <30 cm position error.
6	Maximize User Safety	Braking distance at max speed (m)	Met	Prototype: 5 cm stop distance, 0.2 s stop time, stable during emergency stop. Final design adds dedicated E-stop mushroom button and motor braking at higher operating speeds.
7	Maximize Climable Incline	Max grade angle (degrees)	Partially met	Prototype climbed 10 degrees before motor stall. Limited by motor torque, not traction. Final design uses 50:1 gear ratio (vs prototype 30:1) targeting 15+ degrees. Campus max grade is ~12 degrees.
8	Minimize Modular Changeover Time	Time to swap (minutes)	Met	Prototype: 3 minutes, 1 tool, 8 steps. Final design: dovetail rail + spring-loaded pin targets <2 min with 0 tools for locomotion, <5 min for payload (4 bolts with included security driver).

Appendix A: Concept Development

Concepts Kept

- Concept 1: Fully Autonomous Delivery Robot (Sidewalk Navigation) - Core system enabling campus-wide operation
- Concept 2: Tank Treads - Maximizes traction and improves incline climbing
- Concept 3: Hybrid Wheel System (switchable summer/winter) - All-season functionality
- Concept 4: LiDAR Navigation System - Improves navigation accuracy and obstacle detection
- Concept 9: Speed Limiting System - Improves safety, aligns with constraints

Concepts Dropped

- Concept 5: Chains/Spiked Wheels - Not applicable for London, ON weather conditions
- Concept 6: Food Delivery System (standalone) - Too broad, did not align with evolved need statement
- Concept 7: Dovetail Lock Mechanism for Chassis (remote-controlled) - Too expensive, team lacked expertise
- Concept 8: Rocker-Bogie Setup - Too complex for need statement, does not address winter weather
- Concept 10: Tilt Wheels Forward - Treads found to be superior for traction
- Concept 11: Thermally Insulated Food Storage - Does not align with evolved modular need statement
- Concept 12: Waterproof Casing - Not a separate constraint; addressed by IP65 electronics enclosure
- Concept 13: Thermal Heating System - Internal components generate sufficient heat
- Concept 14: Softer Suspension - Obstacle detection replaces need for suspension-based climbing

Weighted Scoring Summary

Using pairwise comparison (AHP), the team established normalized objective weights: Lifespan (0.25), Traction (0.23), Energy Efficiency (0.18), Minimize Delivery Time (0.16), Weight (0.09), Cost (0.06), Appeal (0.02). The track-based locomotion module scored highest in traction (the second-highest weighted objective), validating its inclusion alongside wheels.

Appendix B: Detailed Test Data

This appendix provides supporting evidence for each test summarized in Table 3. Raw data, calculations, and photographic documentation are included for each test.

Test 1: Maximum Climbable Incline

Setup: Robot placed at base of adjustable ramp. Incline increased in 2-degree increments. Load: unloaded prototype weight.

Raw Data:

- 2 degrees: Pass (no slip, no stall)
- 4 degrees: Pass
- 6 degrees: Pass
- 8 degrees: Pass
- 10 degrees: Pass (no slipping observed)
- 12 degrees: Fail (motor stall, wheels did not slip)

Conclusion: Failure mode was motor torque limitation, not traction. Final design addresses this with 50:1 gear ratio (vs prototype 30:1).

[Insert photo: Robot on ramp at 10-degree incline]

[Insert photo: Angle measurement tool showing incline]

Test 2: Sensor / Obstacle Detection

Setup: Object placed directly in robot path. Multiple trials run under same starting conditions. Robot started approximately 1m from obstacle.

Raw Data:

- Trial 1: Detected, stopped at 15 cm. No collision.
- Trial 2: Detected, stopped at 14 cm. No collision.
- Trial 3: Detected, stopped at 16 cm. No collision.
- Trial 4: Detected, stopped at 15 cm. No collision.
- Trial 5: Detected, stopped at 13 cm. No collision.

Average stopping distance: 14.6 cm. Success rate: 100% (5/5 trials).

[Insert photo: Test setup showing robot and obstacle]

[Insert photo: Robot stopped in front of obstacle]

Test 3: Battery Consumption

Setup: Robot run continuously for 5 minutes on flat, consistent surface. Batteries: 2x 18650 (3200 mAh each), fully charged.

Raw Data and Calculation:

- Initial battery voltage: 4.20 V
- Final battery voltage: 4.12 V
- Voltage drop: 0.08 V
- Average speed: 0.75 m/s
- Distance in 5 min: $0.75 \text{ m/s} \times 300 \text{ s} = 225 \text{ m}$
- Battery capacity: $2 \times 3.2 \text{ Ah} \times 3.7 \text{ V} = 23.68 \text{ Wh}$
- Energy used (estimated from voltage drop): $0.08/4.2 \times 23.68 = 0.451 \text{ Wh}$

- Energy per km: $0.451 \text{ Wh} / 0.225 \text{ km} = 2.28 \text{ Wh/km}$ (prototype, unloaded)

Note: Final design uses 24V 30Ah LiFePO4 (720 Wh). At projected 20 Wh/km under full payload, estimated range is 36 km theoretical, conservatively 12+ km.

[Insert photo: Battery voltage reading before test]

[Insert photo: Battery voltage reading after test]

Test 4: Terrain Test (Wheels)

Setup: Robot operated on gravel and sand surfaces using wheel configuration. Same load and speed conditions as other tests.

Observations:

- Gravel surface: Wheels experienced visible slipping. Robot unable to maintain forward movement.
- Sand surface: Wheels dug into surface. No forward progress achieved.

Result: FAIL. Wheels are not suitable for loose terrain. This confirmed the need for the track locomotion module.

[Insert photo/screenshot: Wheels slipping on gravel]

Test 5: Terrain Test (Tracks)

Setup: Same surfaces and conditions as Test 4, with track configuration installed.

Observations:

- Gravel surface: No slipping. Robot moved forward smoothly.
- Sand surface: No slipping. Robot traversed successfully.

Result: PASS. Tracks provided superior traction on all loose surfaces tested. Validates dual-locomotion modular design approach.

[Insert photo/screenshot: Tracks successfully traversing gravel]

[Insert photo: Side-by-side comparison of wheel vs track on same surface]

Test 6: Emergency Stop

Setup: Robot run at fixed speed on flat surface. Power cut to simulate emergency stop. Stopping distance and time measured. Stability observed via video recording.

Raw Data:

- Trial 1: Stop distance 5 cm, stop time 0.2 s, stable
- Trial 2: Stop distance 4 cm, stop time 0.2 s, stable
- Trial 3: Stop distance 5 cm, stop time 0.2 s, stable

Average: 4.7 cm stopping distance, 0.2 s stopping time. No tipping or instability in any trial.

[Insert screenshot: Video frame showing robot at moment of stop]

Test 7: Module Switch Time

Setup: Full track-to-wheel changeover timed. Operator followed written procedure. Tools required: 1 (screwdriver for prototype fasteners).

Raw Data:

- Trial 1: 3 min 15 sec, 8 steps
- Trial 2: 2 min 50 sec, 8 steps
- Trial 3: 3 min 05 sec, 8 steps

Average: 3 min 3 sec. All trials used 1 tool and 8 steps.

Final design improvement: Dovetail rail with spring-loaded locking pin eliminates tools and reduces steps to 2-3, targeting under 2 minutes.

[Insert photo: Mid-swap showing module partially removed]

[Insert photo: Tools used for prototype swap]

Appendix C: CAD Model

Complete Onshape CAD model:

[Insert Onshape share link here]

The model includes all custom parts, both locomotion module assemblies, and three payload module examples.

Appendix D: Arduino Source Code

[Insert GitHub or repository link here]

Repository includes main.ino, navigation.cpp/h, obstacle.cpp/h, motors.cpp/h, sensors.cpp/h, and route configuration utility.